

# An Integrated Single-Nucleus Atlas Resolves Cell-Type-Specific Programs and Molecular Subtypes in Alzheimer's Disease

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**Abstract**

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## Abstract

Interindividual heterogeneity in Alzheimer's disease (AD) remains poorly understood, as disparate single-cell studies leave it unclear whether findings reflect shared architecture or dataset-specific idiosyncrasies. Here, we present panAD, a transcriptomic atlas of >3 million nuclei from 791 individuals across 13 studies, spanning AD, mild cognitive impairment, and cognitively normal aging. AD converges on a reproducible, cell-type-specific molecular architecture: co-expression modules track neuropathology and cognitive decline; GWAS risk genes act predominantly as downstream targets of transcription factor hubs such as microglial *SPI1*; intercellular communication is remodeled with disease stage; and sex differences concentrate in microglial immune-activation programs. To model patient-level transcriptomic heterogeneity, we developed the Multi-seed Optimization of Neural Embeddings for subTyping (MONET) framework, in which a masked variational autoencoder applied to covariate-adjusted, multi-cell-type profiles resolves four subtypes (Metal-Ion Stress, Neuroinflammatory, Synaptic Integrity, and Tissue Remodeling) that dissociate neuropathological burden from cognitive impairment and nominate predominantly non-overlapping candidate therapeutics. Finally, Stellar Atlas provides an AI-native conversational interface to the atlas.





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Research presentation | September 2026

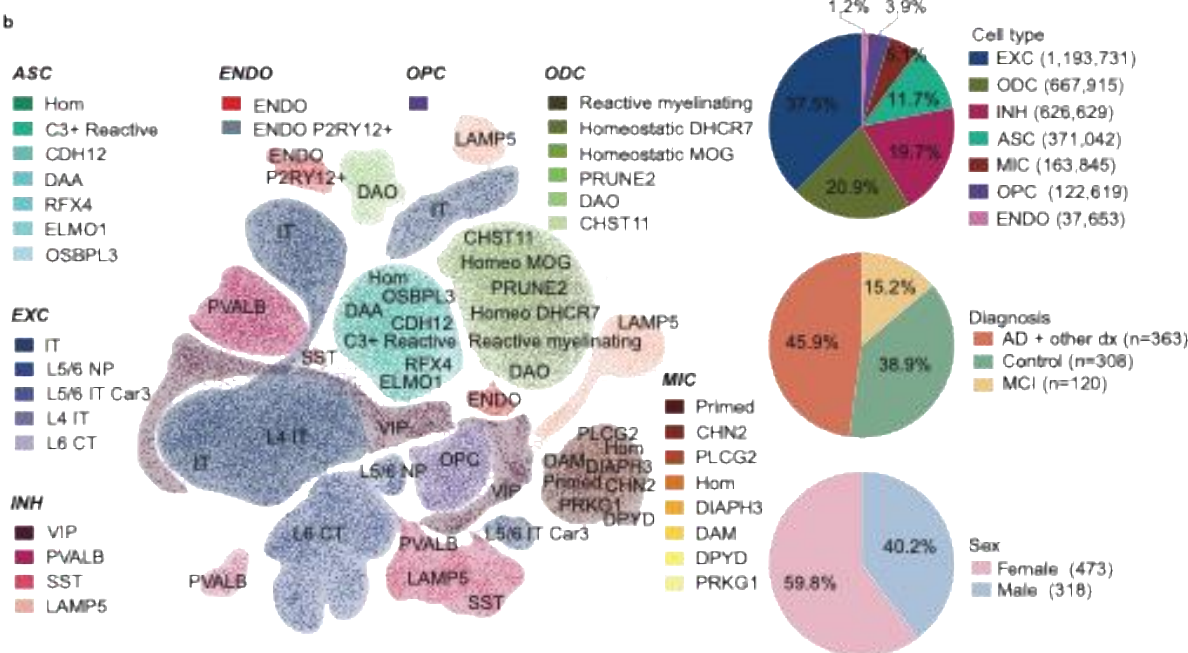
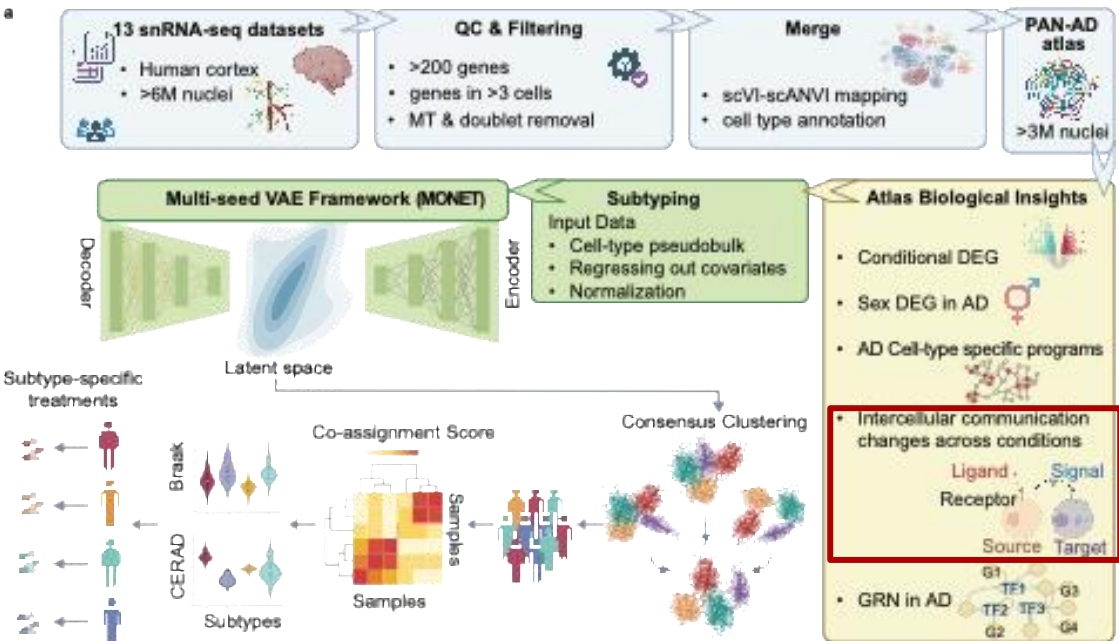
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Charlie Dunlop School of Biological Sciences  
Swarup Lab

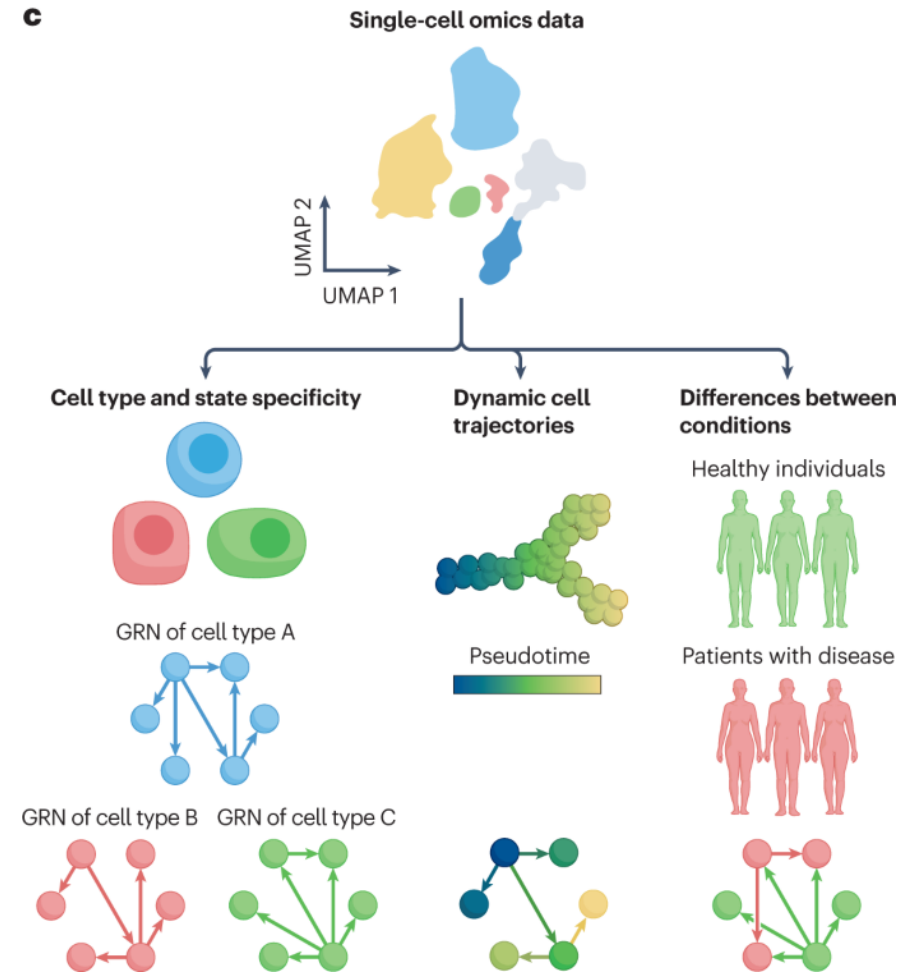
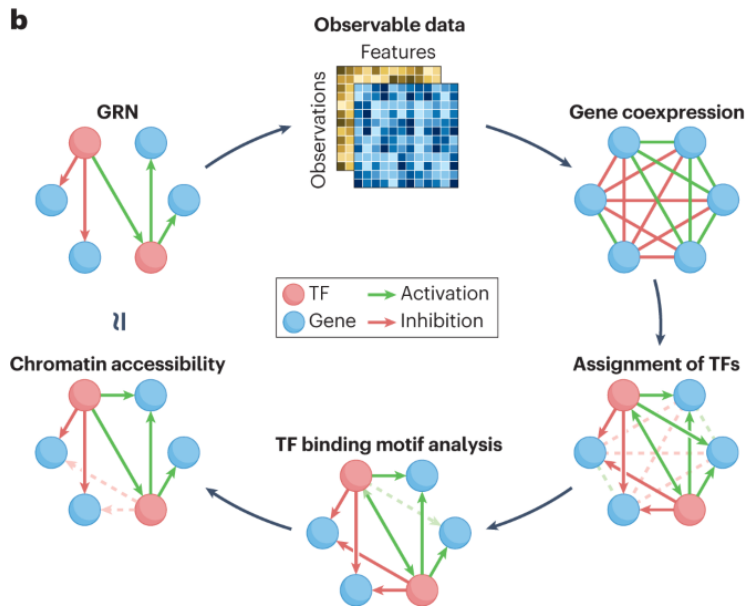
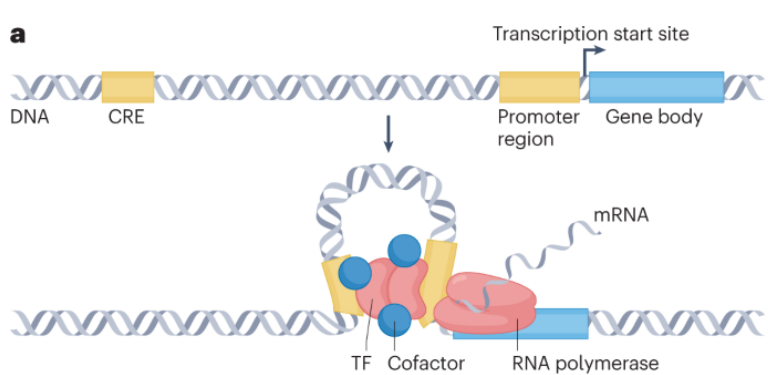


# **STUDY OVERVIEW**

# Study overview



# Principles of Gene Regulatory Networks (GRNs)



# pySCENIC: An Established Method for GRN Inference



## A scalable SCENIC workflow for single-cell gene regulatory network analysis

Bram Van de Sande<sup>1,2,8</sup>, Christopher Flerin<sup>1,2,8</sup>, Kristofer Davie<sup>1</sup>,  
Maxime De Waegeneer<sup>1,2</sup>, Gert Hulselmans<sup>1,2</sup>, Sara Aibar<sup>1,2</sup>, Ruth Seurinck<sup>3,4</sup>,  
Wouter Saelens<sup>3,4</sup>, Robrecht Cannoodt<sup>3,4,5</sup>, Quentin Rouchon<sup>3,4</sup>, Toni Verbeiren<sup>6,7</sup>,  
Dries De Maeyer<sup>6</sup>, Joke Reumers<sup>6</sup>, Yvan Saeys<sup>3,4</sup> and Stein Aerts<sup>1,2</sup>✉

This protocol explains how to perform a fast SCENIC analysis alongside standard best practices steps on single-cell RNA-sequencing data using software containers and Nextflow pipelines. SCENIC reconstructs regulons (i.e., transcription factors and their target genes) assesses the activity of these discovered regulons in individual cells and uses these cellular activity patterns to find meaningful clusters of cells. Here we present an improved version of SCENIC with several advances. SCENIC has been refactored and reimplemented in Python (pySCENIC), resulting in a tenfold increase in speed, and has been packaged into containers for ease of use. It is now also possible to use epigenomic track databases, as well as motifs, to refine regulons. In this protocol, we explain the different steps of SCENIC: the workflow starts from the count matrix depicting the gene abundances for all cells and consists of three stages. First, coexpression modules are inferred using a regression per-target approach (GRNBoost2). Next, the indirect targets are pruned from these modules using cis-regulatory motif discovery (cisTarget). Lastly, the activity of these regulons is quantified via an enrichment score for the regulon's target genes (AUCell). Nonlinear projection methods can be used to display visual groupings of cells based on the cellular activity patterns of these regulons. The results can be exported as a loom file and visualized in the SCoPe web application. This protocol is illustrated on two use cases: a peripheral blood mononuclear cell data set and a panel of single-cell RNA-sequencing cancer experiments. For a data set of 10,000 genes and 50,000 cells, the pipeline runs in <2 h.

# Incorporating Biological Priors in GRN Inference



## The Problem(s)

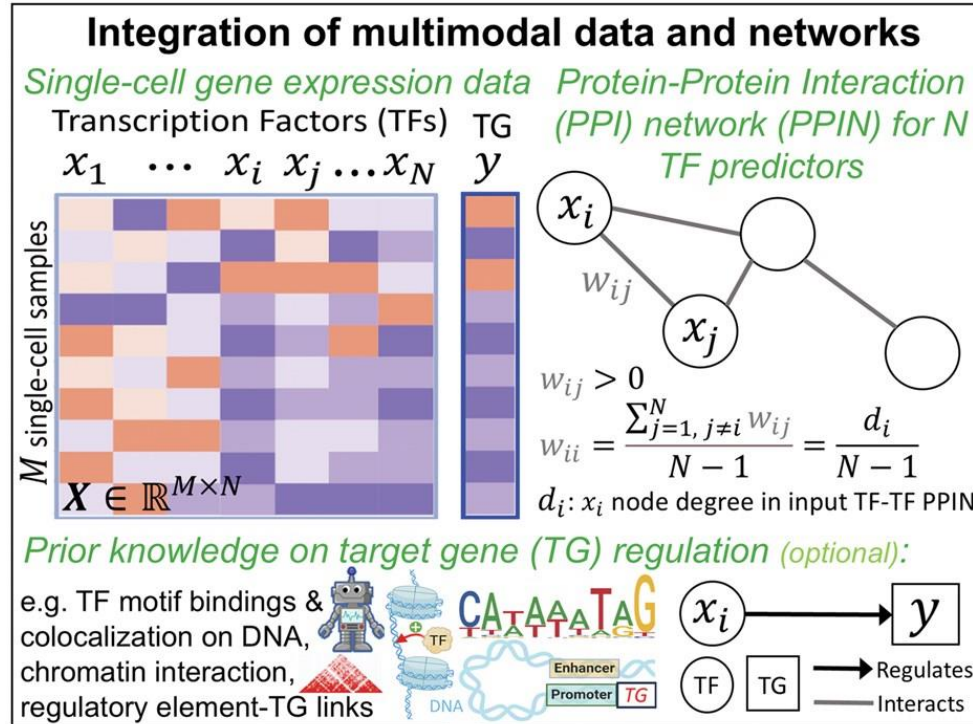
- Gene expression space is high-dimensional
- Noisy and sparse
- Assuming TFs act independently
- No regulatory directionality

## The solution(s)

- Biological mechanisms happen through coordinated TF programs
- Reduce spurious associations
- Use validated knowledge for biological significance
  - “Guiding the regression”
  - Prioritize TFs with stronger connection, and nominate new links
- Enable cell-type- and condition specific regulation

**GRN inference is less about discovering networks and more about constraining biological hypotheses under uncertainty.**

# NetREm: Network Regression Embeddings



## Step 1: Network regularized regression definition

$$c^* = \operatorname{argmin}_c \left\{ \frac{1}{2M} \|y - Xc\|^2 + \alpha \|c\|_1 + \frac{\beta}{2} \sum_{i=1}^N \sum_{j=1}^N \left( w_{ij} \left( \frac{c_i}{\sqrt{d_i}} - \frac{c_j}{\sqrt{d_j}} \right)^2 \right) \right\}$$

$c^* \in \mathbb{R}^N$ : regression coefficients of  $N$  candidate TFs for predicting TG expression levels  $y$  for TF-TG regulation

Hyperparameters:  $\beta > 0$ : network-constrained prior  $\alpha \geq 0$ : sparsity-prior

## Step 2: Gene embeddings from network regularized regression

$$c^* = \operatorname{argmin}_c \left\{ \frac{1}{2N} \|\tilde{y} - \tilde{X}c\|^2 + \alpha \|c\|_1 \right\}$$

where

$$\tilde{X} = \sqrt{N} \Sigma_{\text{trunc}}^{\frac{1}{2}} U^T \in \mathbb{R}^{N \times N}$$

$$\tilde{y} = \frac{\sqrt{N}}{M} \Sigma_{\text{trunc}}^{-\frac{1}{2}} U^T X^T y \in \mathbb{R}^N$$

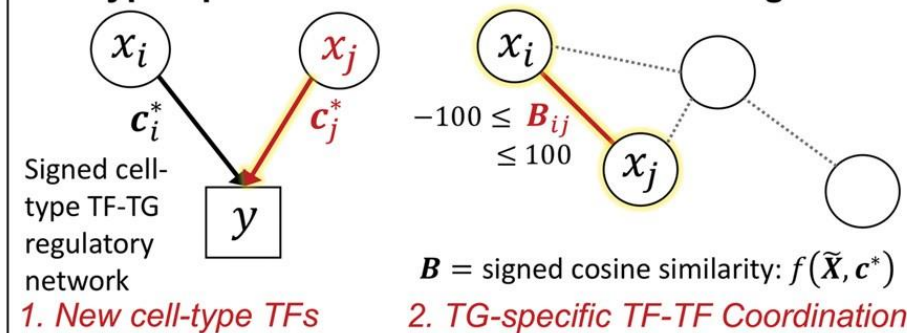
Singular Value Decomposition (SVD) of the  $E \in \mathbb{R}^{N \times N}$  matrix:

$$E = U \Sigma U^T = \frac{X^T X}{M} + \beta D^T (W \odot V) D$$

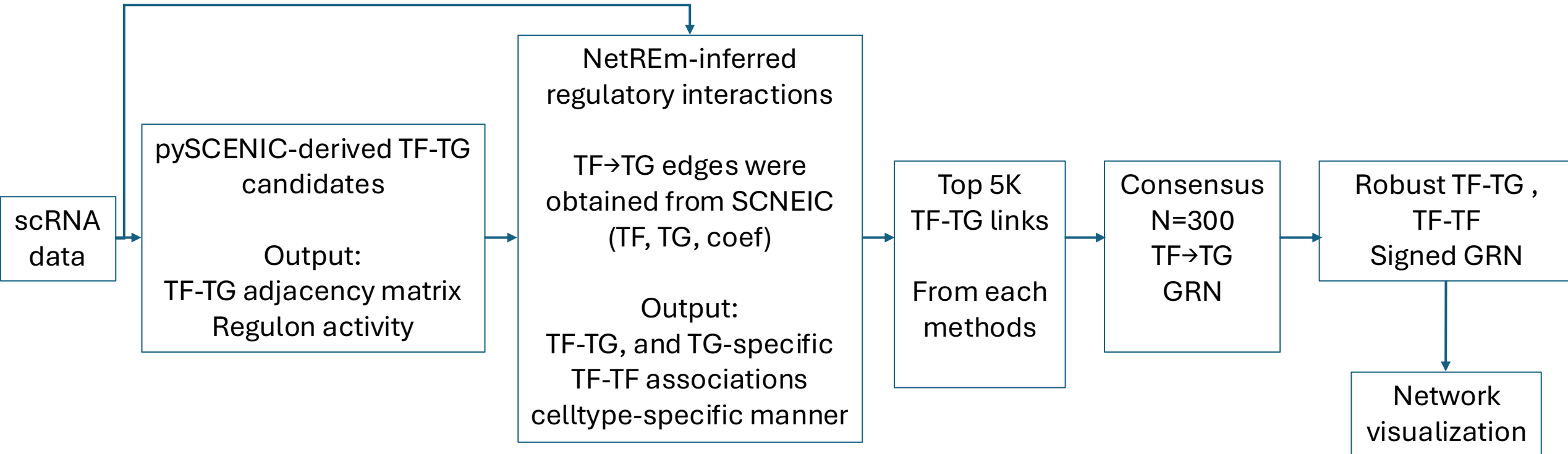
$D$ : diagonal matrix of TF node degrees  $d$

$V = N \cdot I - \mathbf{1}\mathbf{1}^T$   $W$ : matrix of TF-TF PPIN edge weights  $w > 0$

## Cell-type-specific TF coordination for TG regulation

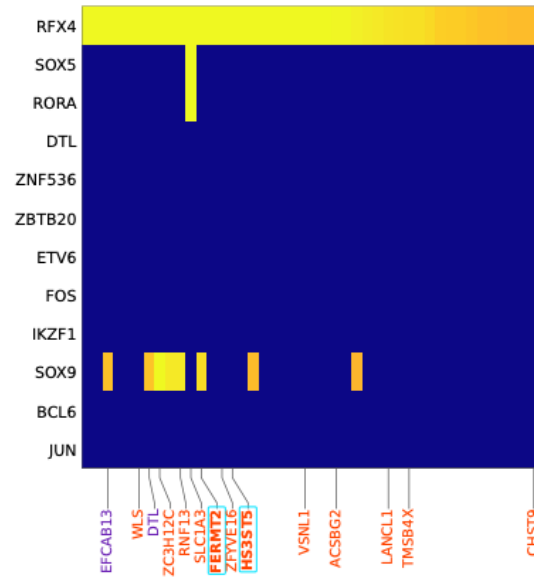


# Consensus GRN Reconstruction

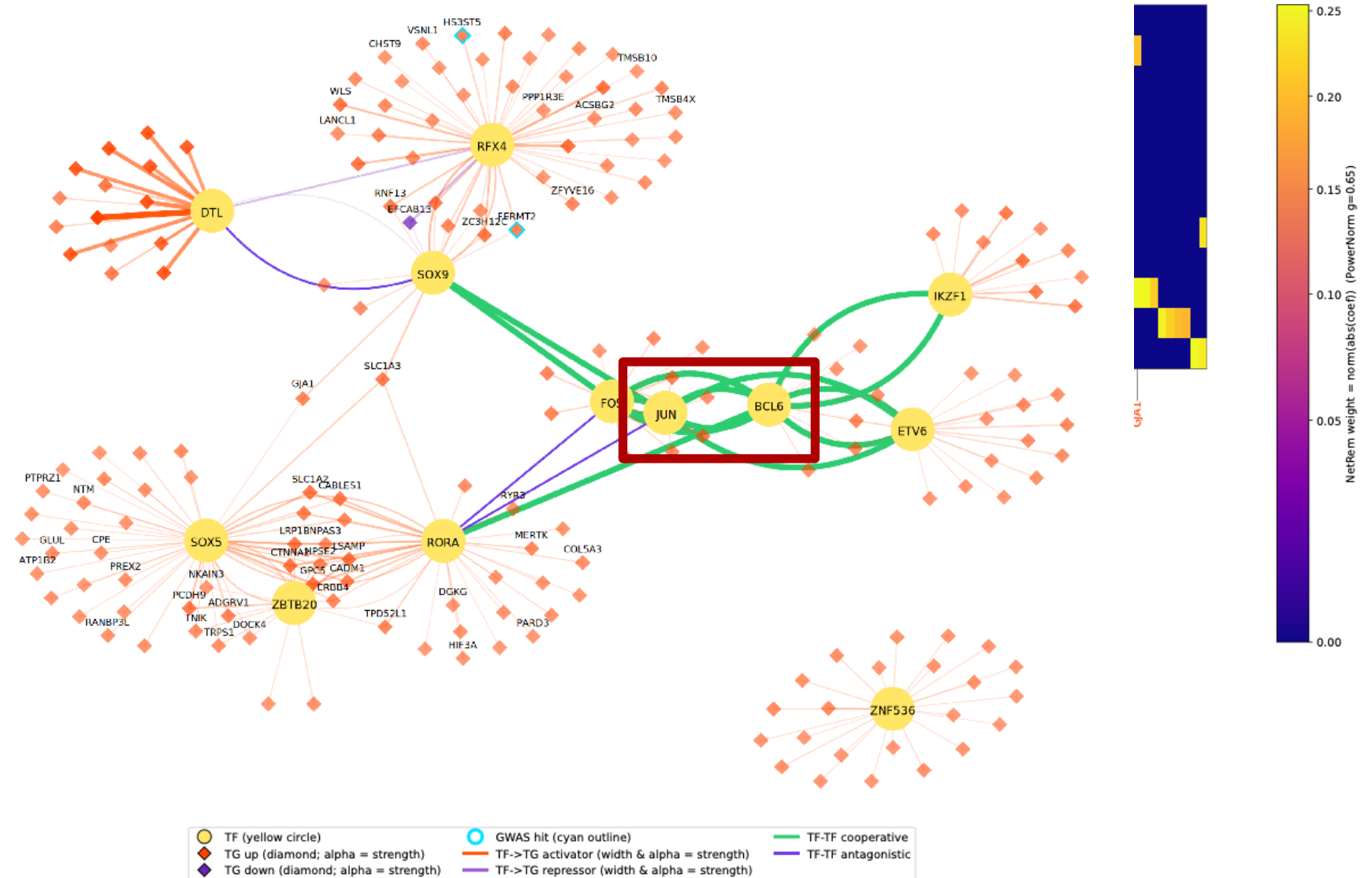


# Astrocyte-Specific Regulatory Network in AD

TF-TF + TF->TG network - AST



NetRem identified **JUN-BCL6**, a TF-TF interaction previously characterized in immune cell differentiation<sup>1</sup>, as a candidate regulatory relationship in AD.

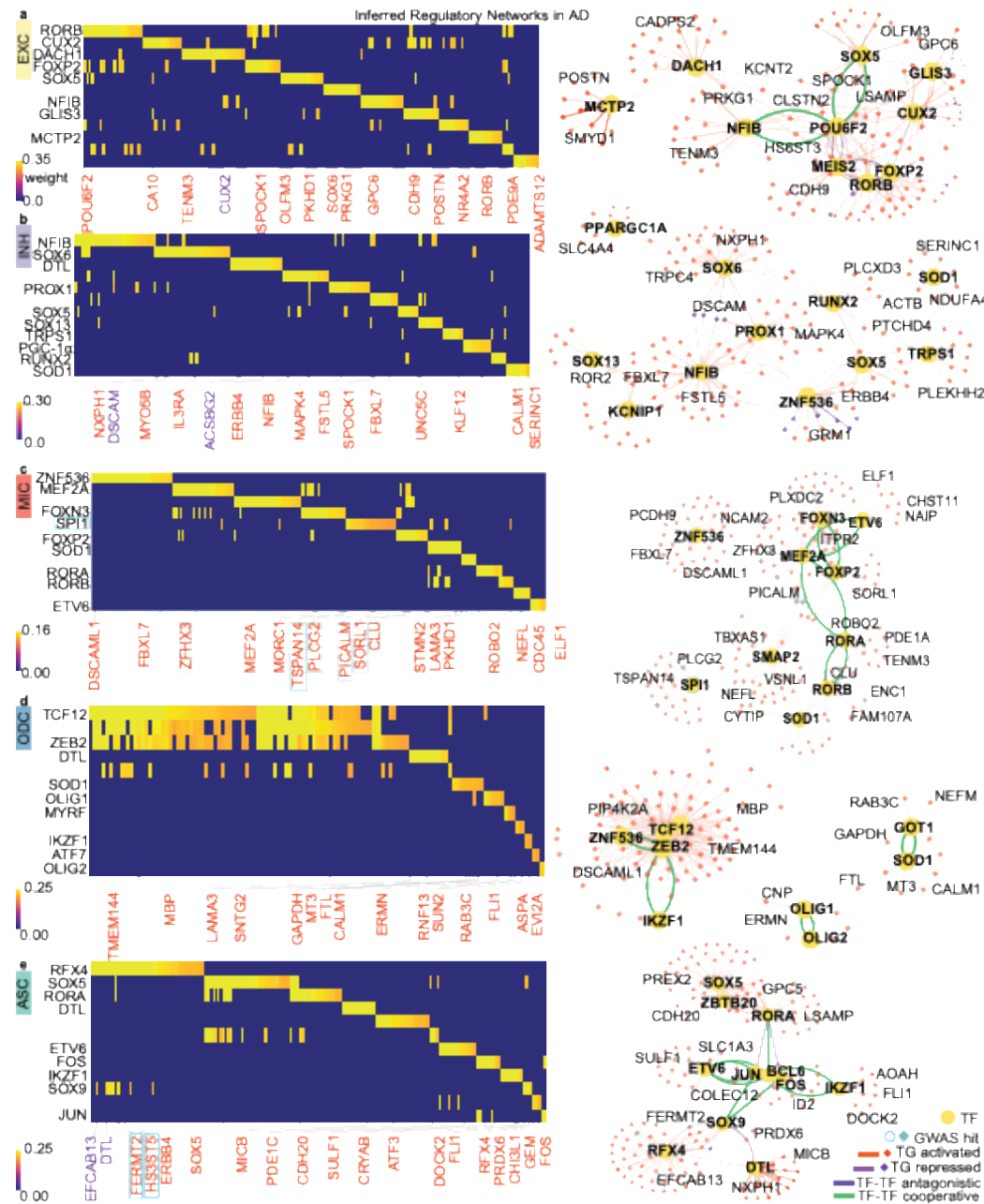


<sup>1</sup>Vasanwala FH, Kusam S, Toney LM, Dent AL. Repression of AP-1 function: a mechanism for the regulation of Blimp-1 expression and B lymphocyte differentiation by the B cell lymphoma-6 protooncogene. J Immunol. 2002 Aug 15;169(4):1922-9. doi: 10.4049/jimmunol.169.4.1922. PMID: 12165517.

Combining pySCENIC and NetREm revealed Cell-type-specific transcription factor networks in AD, including distinct regulatory relationships in neurons, microglia, oligodendrocytes, and astrocytes.

NetREm let us move beyond asking which genes a transcription factor regulates, to asking which transcription factors work together revealing cooperative and antagonistic TF-TF relationships underlying AD-associated transcriptional programs in each cell type.

Rahimzadeh, N. *et al.* An Integrated Single-Nucleus Atlas Resolves Cell-Type-Specific Programs and Molecular Subtypes in Alzheimer's Disease. *bioRxiv*, 2026.2009.2003.747935, doi:10.64898/2026.09.03.747935 (2026).



[https://swaruplab.bio.uci.edu/panad\\_atlas/intro](https://swaruplab.bio.uci.edu/panad_atlas/intro)

← → ↺

swaruplab.bio.uci.edu/panad\_atlas/intro

☆ Z | ⬇️

School ⋮

panAD Atlas 3,068,923 cells · 51 modules · 322 comparisons

Share Dark Operon

Intro Chat UMAP Expression DE: AD/MCI/Ctrl DE: Sexes DE: Subtypes Enrichment Network CellChat Supplementary materials

## Pan-AD Atlas

A harmonized single-nucleus RNA-sequencing atlas of the ageing and Alzheimer's human cortex: **3,183,434 nuclei from 791 individuals** spanning Alzheimer's disease, mild cognitive impairment and cognitively normal controls, integrated from 13 published cortical datasets into one cell-type-annotated reference. A masked variational autoencoder (MONET) resolves four reproducible molecular subtypes of AD (AD1-AD4), and every donor carries neuropathological and cognitive covariates — Braak stage, CERAD plaque score, MMSE — so the biology can be read against disease severity, not just diagnosis. On top of that scaffold sit differential expression, sex-stratified analysis, hdWGCNA co-expression networks, cell-cell communication (CellChat), and per-cell-type gene regulatory networks (NetRem + pySCENIC) — all queryable interactively here, in natural language through the Chat tab, or downloadable as the original supplementary files.

|                                                         |                                               |                                                    |                                                 |                                               |
|---------------------------------------------------------|-----------------------------------------------|----------------------------------------------------|-------------------------------------------------|-----------------------------------------------|
| NUCLEI SERVED<br><b>3,068,923</b><br>of 3,183,434 total | DONOR IDS<br><b>829</b><br>Control · MCI · AD | GENES<br><b>25,251</b><br>with measured expression | DE COMPARISONS<br><b>322</b><br>MAST + Wilcoxon | HDWGCNA MODULES<br><b>51</b><br>per cell type |
|---------------------------------------------------------|-----------------------------------------------|----------------------------------------------------|-------------------------------------------------|-----------------------------------------------|

Counts above are live from the database and describe what the interactive views serve — the AD, MCI and Control diagnostic groups only. The manuscript's headline figures, 791 individuals and 3,183,434 nuclei, describe the full integrated dataset as analysed in the paper.

### Not just a viewer — an analysis environment

Most public single-cell browsers stop at a coloured UMAP and a gene search bar. This atlas was built to go further: lasso any subset of cells, run a Welch-t differential test on the spot, push the up-genes to live enrichment (EnrichR), pivot into hub-gene co-expression networks per cell type, and ask the same database natural-language questions through the Chat tab — which has also read the manuscript, so "what does the paper say about SPI1" and "how many microglia come from Frequent-plaque donors" are answered side by side, with page citations for the former and live counts for the latter. Every plot exports as PNG; every table exports as CSV.